

Yiannis Dimitrakopoulos

Software Engineer · Applied AI

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Profile

Software engineering student at Codam College (42 Network) building toward applied AI engineering roles. Hands-on experience implementing constrained decoding, hybrid retrieval and knowledge-graph systems for LLMs from first principles, and measuring where they actually work. 1st place at the Artifec × Codam Hackathon 2026.

Skills

Programming Languages
Python, C, JavaScript, TypeScript

AI / ML & LLM Systems
Constrained decoding (FSM, logit masking), hybrid retrieval (BM25 + dense, RRF fusion), knowledge graphs, entity resolution, retrieval evaluation (recall@k), open-source model serving (vLLM, Qwen3, Qianfan-OCR)

Web Development
Vue.js, Nuxt, Tailwind CSS, Node.js, Express.js, Fastify, REST APIs

Databases
PostgreSQL (Supabase), Neo4j, MongoDB

Tools
Git/GitHub, Bash (Unix shell)

Projects

Constrained-GraphRAG
Knowledge-Graph Retrieval with Constrained Extraction
2026 · github.com/DimYiannis/Constrained-GraphRag

- Python pipeline in which Qwen3-0.6B with FSM-based constrained decoding extracts entities and relationships from the vLLM codebase into a Neo4j knowledge graph, using a closed relation schema.
- Designed the graph data model and entity resolution that merges duplicate entities into one.
- Evaluated graph-augmented against hybrid-only retrieval (recall@k with matched and random controls) to measure where the graph helped and where it didn't.

Call Me Maybe
Constrained Decoding Engine for LLMs
2026

- Built a constrained-decoding pipeline from scratch for Qwen3-0.6B: manual logit masking, an FSM tracking valid JSON states and a generation loop on raw logits, with no ML frameworks. Implements the mechanism used inside vLLM/SGLang for structured output.

RAG Against The Machine
Hybrid Retrieval System
2026

- Built a hybrid BM25 + dense retrieval pipeline with Reciprocal Rank Fusion (RRF), a caching layer, and two chunking strategies: AST-based for Python source, plain chunking for markdown/text.

Education

Codam College
Amsterdam
10/2025 – Present

- Intensive, project-based curriculum focused on **C, Python, algorithms and collaborative software engineering**.
- Every project is **peer-evaluated** through code review and defence sessions with other students.

Aristotle University of Thessaloniki (AUTH)
Bachelor's Degree, Physical Education & Sport Science
10/2017 – 07/2021

Hackathons & Awards

Artifec × Codam Hackathon — 1st Place
Verumly
10/2026

- Won first place out of seven teams. Verumly is a privacy-first digital receipt platform: the till issues a signed receipt as a QR code that customers save to their wallet with one scan.
- Digitised paper receipts with Baidu's vision-language model Qianfan-OCR, pinned to a fixed revision and served with vLLM on a GPU behind an OpenAI-compatible API.
- TypeScript monorepo with Nuxt 4 apps, Fastify APIs and Postgres (Supabase).

Build Weekend Amsterdam — Young Creators, Finalist
11/2025

- Built "ema.", a voice assistant guiding older adults through daily travel, using n8n, ElevenLabs and Twilio.

Lely Hackathon — Embedded Programming
11/2025

- Programmed mBot robots in Python in a one-day embedded-systems challenge.

Professional Experience

Front Desk Staff
Hertz Autohellas
04/2024 – 10/2024

- Handled customer inquiries, concerns and requests with accuracy and care.
- Communicated daily with customers and colleagues in a fast-paced team.

Greek Army
Secretary Assistant

- Twelve-month obligatory service as secretary assistant and assistant for validation of expenditures.

Languages

English — Proficient (Certificate of Proficiency in English, ECPE)
Greek — Native